Heart-Disease Risk API: an end-to-end MLOps pipeline

Machine Learning Operations (AIMLCZG523) · Assignment 1 · Rajesh Patel (BITS ID: 2025AE05357)

Repository: https://github.com/2025AE05357-sys/heart-disease-mlops

> Educational project. The model is not a medical device and its output is not medical advice.

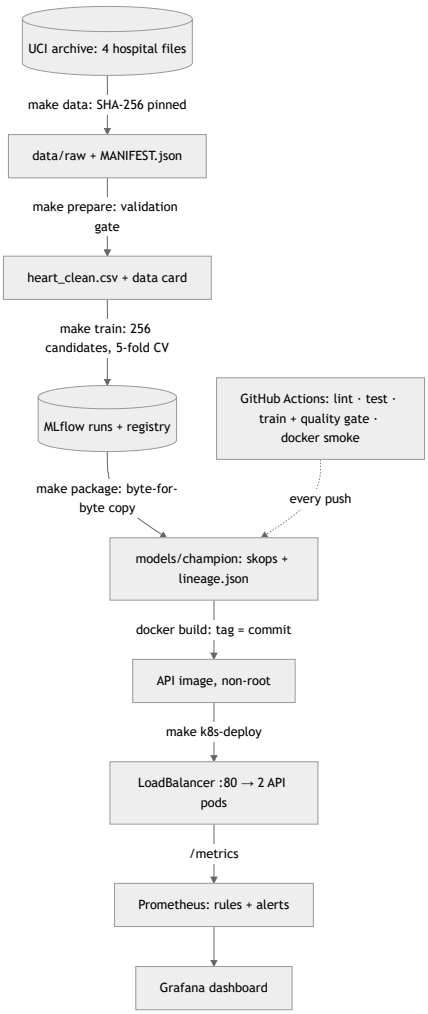
1. Introduction

The task was to build a classifier that predicts heart disease from routine clinical measurements and to take it all the way to a monitored API on Kubernetes, using MLOps practice at every step: reproducible data, tracked experiments, packaged models, CI, containers, deployment manifests and monitoring.

I gave the project one organising principle, taken from my work on GxP-regulated clinical data systems: **every prediction must be traceable.** For any answer the API gives, it must be possible to say which model version produced it, which training run built that model, from which code commit, and from which exact bytes of data. This principle decided many of the design choices below, and each section points out where it shows up.

Objective	Where it is met
Reproducible data with provenance	Section 3: checksum-pinned download, validation gate, data card
At least two tuned, cross-validated models	Section 4: logistic regression, random forest, gradient boosting
Every run tracked	Section 5: MLflow parent, family and candidate runs; registry aliases
A reusable, reproducible model artifact	Section 6: MLflow model in skops format with `lineage.json`
CI that fails loudly	Section 7: GitHub Actions with lint, tests, training and a quality gate
A containerised API	Section 8: FastAPI, non-root image, audit log
Deployment on Kubernetes	Section 9: kustomize manifests, LoadBalancer, self-healing
Monitoring	Section 10: Prometheus and Grafana, with a drift demonstration

Architecture



2. Setup and installation

Item	Primary environment	Clean-machine alternative
Operating system	Windows 11 with WSL2 Ubuntu 24.04	BITS lab VM (Rocky Linux 9.5)
Python	3.12, one virtual environment, fully pinned lock file	same
Containers and cluster	Docker Desktop with its single-node Kubernetes (v1.36)	Docker Engine and minikube
Experiment store	MLflow on SQLite under `~/.heartrisk/mlflow`	same

`scripts/vm_bootstrap.sh` sets up either environment idempotently: system packages, Docker access, the Kubernetes context, and the virtual environment from `requirements.txt`. Dependencies are declared in `requirements.in` and resolved with `uv` into a fully pinned `requirements.txt`; the API image uses a second lock, `requirements-serve.txt`, compiled with the training lock as a constraint so that scikit-learn, pandas, NumPy, SciPy and skops are identical at training and serving time.

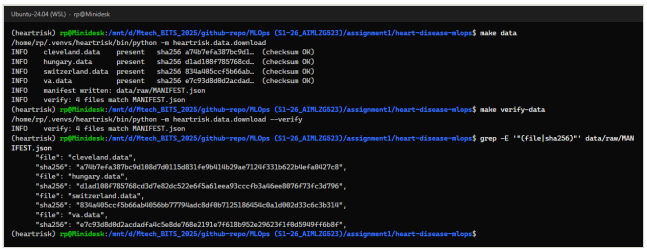
Every stage is one `make` target:

```
make all                                # verify data -> validation gate -> train -> package
make docker-build docker-run smoke      # the API in a container on :8000
make k8s-deploy k8s-smoke               # the API on Kubernetes behind http://localhost
make monitoring-up traffic traffic-shift   # Prometheus :9090, Grafana :3000, drift demo
```

3. Data and exploratory analysis

3.1 Acquisition and provenance

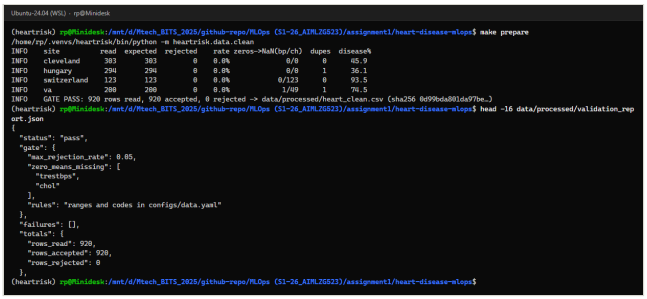
The brief links to the UCI Heart Disease dataset, which is four hospital databases, not one file: Cleveland (303 patients), Hungary (294), Switzerland (123) and VA Long Beach (200), each with the same 14 attributes. `make data` downloads the four files and writes `data/raw/MANIFEST.json` with the URL, size, SHA-256 and retrieval time of each, plus the licence (CC BY 4.0) and citation. Later runs reuse local files that still match the manifest and refuse upstream bytes that differ from it, so the training data cannot change without a visible diff. The raw files are committed.



Download and manifest verification

3.2 Validation gate and cleaning

`make prepare` runs a validation gate built with pandera from `configs/data.yaml`. It parses `?` as missing, and it treats zeros in cholesterol and resting blood pressure as missing (decision D2): no living patient has either value, and the Swiss file records every cholesterol as 0. Every row is then checked against clinical plausibility ranges (for example cholesterol 80–700 mg/dl) and code sets; a failing row goes to `rejected.csv` with its reason instead of being dropped silently, and a site fails the run if its row count changes or more than 5% of its rows are rejected. All 920 real rows pass. Imputation deliberately does not happen here: it belongs in the model pipeline, fitted on training folds only. Each clean row keeps a `record_id` pointing to its line in the raw file, and the generated data card records the SHA-256 of the output. Re-running the gate produces byte-identical files, which CI checks on every push.



Validation gate summary

3.3 Exploratory analysis

The notebook `notebooks/01_eda.ipynb` (executed by `make eda`) compares the four sites and studies Cleveland in depth. Figure 1 shows why decoding the zeros matters: Switzerland looks complete on cholesterol until its zeros are read as "not measured".

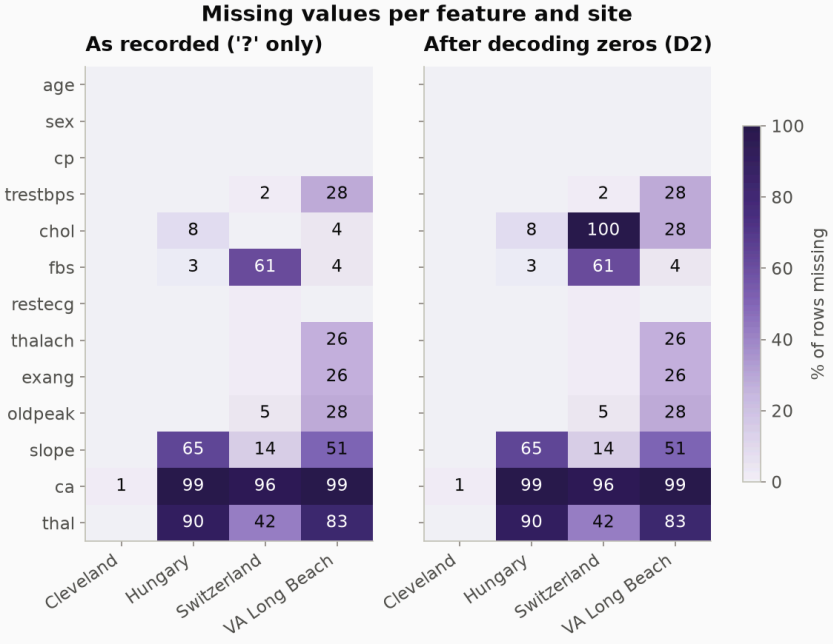


Figure 1: missing values per feature and site, as recorded and after decoding zeros

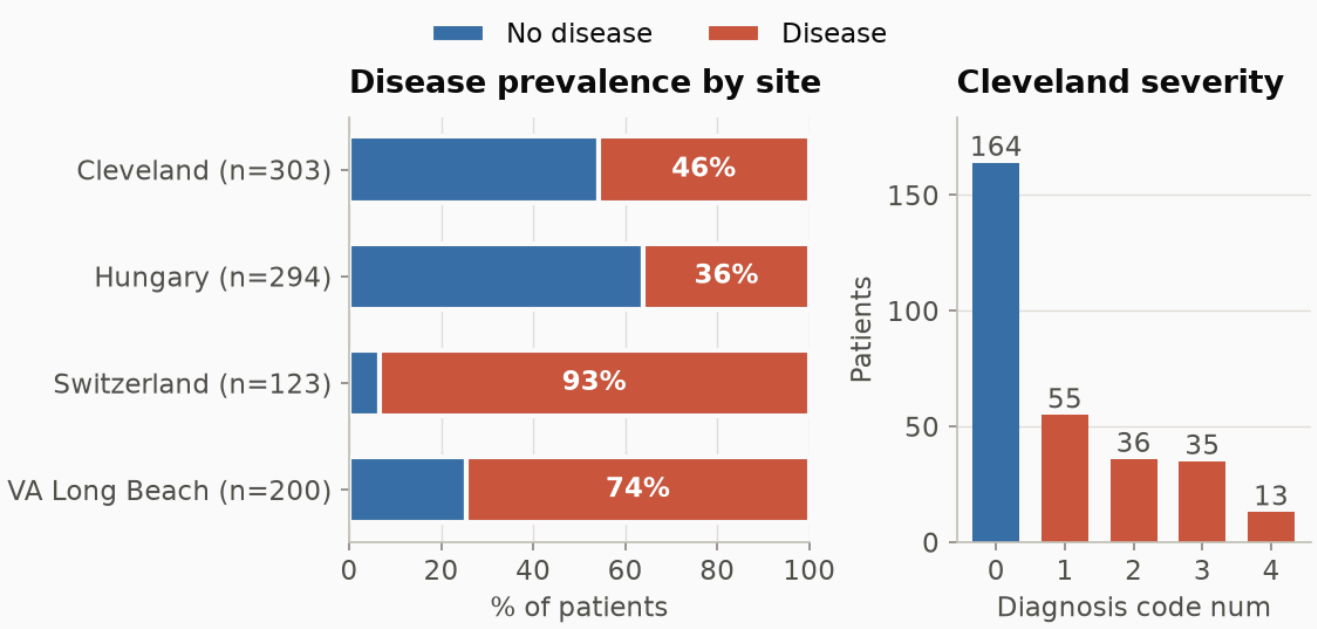


Figure 2: disease prevalence by site, and Cleveland's severity codes mapped to the binary target

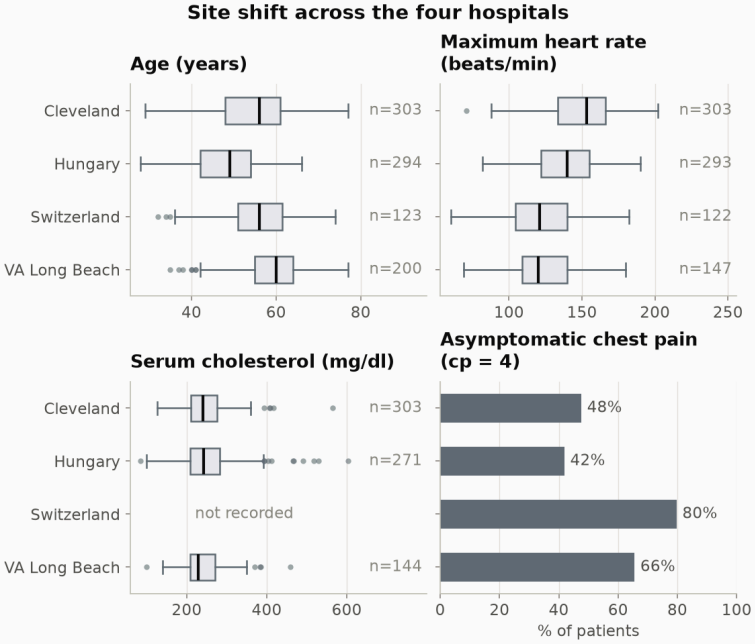


Figure 3: the four hospitals are different populations

#	Finding	Consequence for modelling
1	Cleveland is almost complete (2.0% of rows have a gap); at the other sites 99.5–100% of rows do, and `ca` is missing for 96–99% of patients	Train on Cleveland; impute inside the pipeline; expect weaker external results
2	Prevalence ranges from 36.1% (Hungary) to 93.5% (Switzerland), against 45.9% in Cleveland	Compare sites on ROC-AUC, which does not depend on prevalence
3	Strongest Cleveland signals: vessels on fluoroscopy (Spearman $\rho = +0.49$), exercise angina (+0.43), maximum heart rate (−0.42), ST depression (+0.41); cholesterol is weak (+0.12)	The clinically expected predictors dominate
4	External sites differ: median maximum heart rate 120–121 at Switzerland and VA vs 153 in Cleveland; asymptomatic chest pain 80% in Switzerland vs 48%	Built-in covariate shift, reused for the drift demonstration
5	Cleveland is close to balanced (164 without disease, 139 with)	No resampling; stratified cross-validation

Data scope (decision D1). Pooling all 920 patients would triple the sample, but the site predicts the label (Switzerland is 93.5% positive) and the missing-value pattern reveals the site, so a pooled model could learn *which hospital* instead of physiology. I therefore train and tune on Cleveland only and keep Hungary, Switzerland and VA Long Beach as external validation sets.

4. Features and model development

4.1 Pipeline

Every learned transformation lives inside one scikit-learn pipeline, so it is fitted on training folds only and saved with the model:

Step	Columns	Transformation
Engineered feature (D3)	age, maximum heart rate	Optional `hr_pct_pred`, tuned as a hyperparameter
Continuous	age, resting BP, cholesterol, max heart rate, ST depression	Median imputation, standardisation
Ordinal	vessels on fluoroscopy (`ca`)	Mode imputation, standardisation
Binary	sex, fasting blood sugar, exercise angina	Mode imputation
Nominal	chest pain, resting ECG, ST slope, thallium	Mode imputation, one-hot encoding that ignores unseen categories

The engineered feature is the share of the age-predicted maximum heart rate that the patient reached in the exercise test:

$$\text{hr_pct_pred} = \frac{\text{thalach}}{220 - \text{age}}$$

Notation: thalach = maximum heart rate reached (beats/min); age = age (years); $220 - \text{age}$ = conventional age-predicted maximum heart rate (beats/min); the result is unitless (0.8 means 80% of the predicted maximum).

4.2 Models and tuning

Cleveland's 303 patients are split once, stratified and seeded, into 242 for training and cross-validation and 61 for a locked hold-out set that no tuning step sees. Each model family (D4) is tuned by grid search under stratified 5-fold cross-validation, scoring ROC-AUC, accuracy, precision, recall and F1 and refitting on ROC-AUC: 256 candidates in total (logistic regression 32, random forest 192, gradient boosting 32), 1,280 fits in about four minutes. scikit-learn 1.9 deprecates `LogisticRegression(penalty=…)`, so the L1/L2 choice is expressed as `l1_ratio ∈ {0, 1}`.

For a threshold of 0.5 on the predicted probability:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}$$

Notation: TP = patients with disease predicted as disease; FN = patients with disease predicted as healthy; TN = healthy patients predicted as healthy; FP = healthy patients predicted as disease (all counts of patients).

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Notation: F_1 = harmonic mean of precision and recall (unitless, 0 to 1).

$$\text{ROC-AUC} = \frac{1}{n_1 n_0} \sum_{i:\, y_i = 1} \sum_{j:\, y_j = 0} \mathbf{1}\left[p_i > p_j\right]$$

Notation: n_1, n_0 = numbers of patients with and without disease; y_i = true class of patient i; p_i = predicted probability of disease for patient i; $\mathbf{1}[\cdot]$ = 1 when the condition holds, else 0 (ties count ½). ROC-AUC is the probability that a random diseased patient is ranked above a random healthy one: independent of the threshold and of prevalence.

4.3 Selection and results

The champion is chosen on cross-validation, never on the hold-out set (D5): the best mean CV ROC-AUC wins, but families within 0.01 of the best are treated as tied, because such differences are inside the fold-to-fold spread; among tied families the higher mean CV recall wins, because a missed diagnosis costs more than a false alarm; a remaining tie goes to the simpler model.

Model	CV ROC-AUC	CV recall	Hold-out ROC-AUC	Hold-out recall	Hungary	Switzerland	VA
Logistic regression	**0.909 ± 0.011**	**0.819**	0.950	0.893	**0.895**	0.741	0.698
Random forest	0.904 ± 0.025	0.774	0.948	0.929	0.895	0.771	0.726
Gradient boosting	0.890 ± 0.020	0.792	0.951	0.893	0.841	0.739	0.705

Logistic regression (L1, C = 1, balanced classes) tied with random forest on ROC-AUC and won on recall. The external columns vindicate the rule: gradient boosting had the best hold-out accuracy (0.918) but collapsed at Hungary, where its recall fell to 0.13 against 0.62 for logistic regression. Choosing on the 61-patient hold-out set would have picked the model that transfers worst.

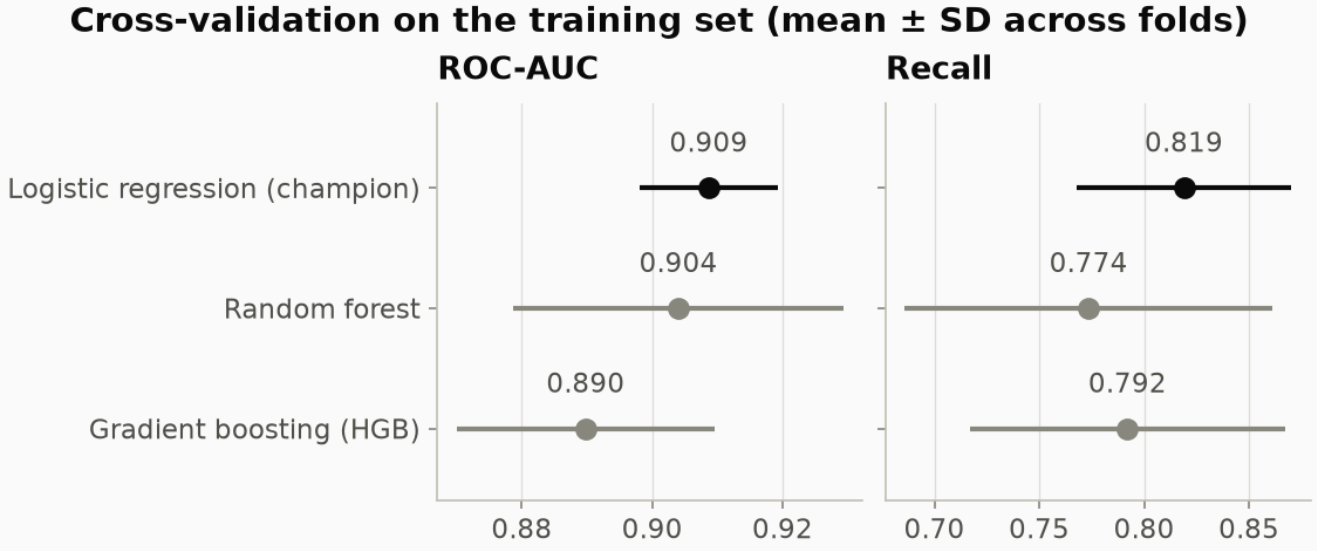


Figure 4: cross-validated ROC-AUC and recall per family

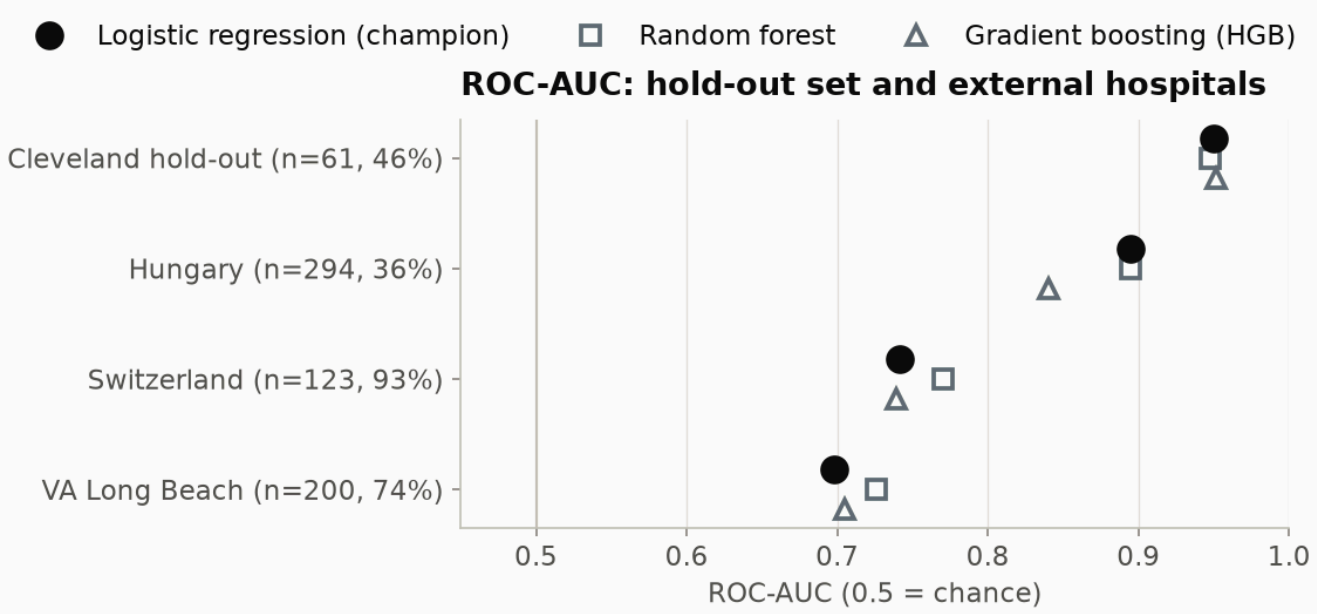


Figure 5: ROC-AUC on the hold-out set and at the three external hospitals

All three models lose discrimination at Switzerland and VA (ROC-AUC 0.70–0.77). These patients are older, reach lower heart rates and lack the fluoroscopy and thallium tests that are the strongest predictors in Cleveland. This is the central limitation of the model, and Section 10 turns it into the monitoring design.

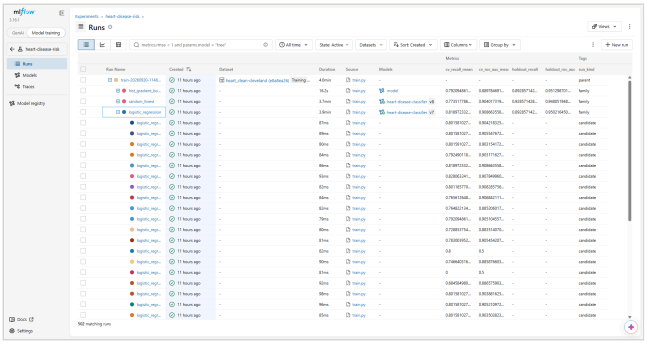
Threshold (D6). The class threshold stays at 0.5; on the hold-out set this gives recall 0.893, and reaching 0.90 would need 0.475, a negligible change that would be fitted to 61 patients. The API additionally returns a risk band (low below 0.3, high from 0.6) as a reading aid.

5. Experiment tracking

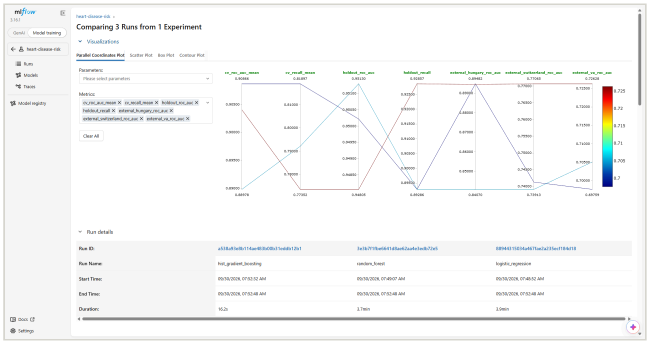
MLflow records every run in a three-level hierarchy: one parent run per `make train`, one child run per model family, and one nested run per hyperparameter candidate (256 per training).

Level	Logged
Parent	Lineage tags (git commit, uncommitted-changes flag, SHA-256 of the clean data and of the raw manifest, library versions); the training dataset with its digest; the config; comparison figures; reports
Family	Best parameters; mean and SD of every CV metric; hold-out and external metrics; ROC, confusion, calibration and importance plots; the full CV table; the fitted pipeline as an MLflow model
Candidate	Parameters; mean and SD of every CV metric; rank

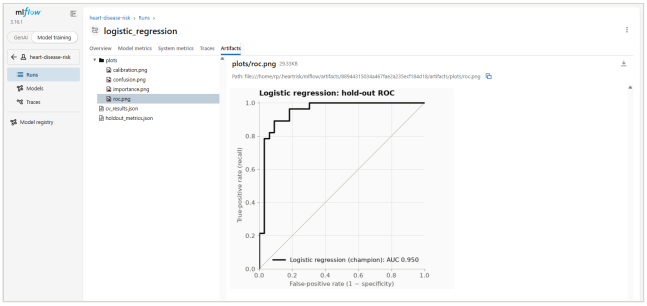
The champion and the runner-up are registered as `heart-disease-classifier` under the aliases `champion` and `challenger` (stages are deprecated in MLflow 3), with the selection reason, data hash and commit as version tags. `make experiments-report` exports the family runs to `reports/experiments.csv`.



MLflow experiment with parent, family and candidate runs



MLflow run comparison



Plots logged inside a family run



Registered model with the champion alias

6. Packaging and reproducibility

`make package` copies the registered champion **byte for byte** into `models/champion/`: an MLflow model directory with the pipeline serialised by skops, its signature, an input example and pinned requirements. skops was chosen over pickle (D7) because loading a pickle can execute arbitrary code from the file, whereas skops only reconstructs types on a trust list. That list is pinned in code; the loader refuses a model whose own declared list goes beyond it, which a test demonstrates.

Before writing anything, packaging verifies two facts and fails otherwise: the clean data still has the hash recorded at training, and the exported file reproduces the hold-out ROC-AUC logged at training time (0.95021645 in both cases for the current model). `lineage.json` then records the registry version, runs, commit, data hashes, rows per split and site, library versions and all metrics, including a slice by sex; `docs/MODEL_CARD.md` is generated from it. The model folder is committed (D8), so the container build needs neither retraining nor access to a registry.

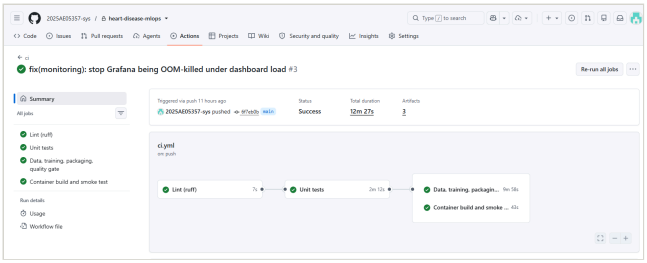
Reproducibility is tested at three levels: retraining with the same seed reproduces the comparison table exactly; re-running the data gate reproduces the clean files byte for byte; and the exported model gives predictions identical to the registry copy.

7. CI/CD and automated tests

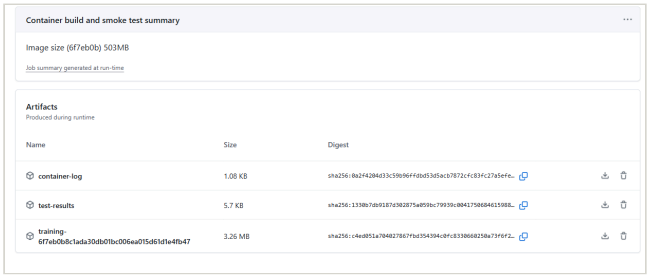
The test suite has 104 tests with 95% line coverage and no network access: hand-written raw rows in each site's format for the data gate, hand-worked metric examples, a real 121-row sample for fast end-to-end training, packaging and API tests, and security tests for the model loader.

Job	Steps	Evidence produced
`lint`	ruff check and format check at the locked version	Annotations on the pull request
`test`	pytest with coverage	JUnit and coverage XML (uploaded even on failure), coverage table in the job summary
`train`	Verify raw data against the manifest; re-run the validation gate and **fail if a single byte of `data/processed` differs from git**; train; package; quality gate (D9: hold-out ROC-AUC ≥ 0.80)	Reports, model, model card and MLflow store as artifacts; metrics table in the job summary
`docker`	Build the image tagged with the commit, run it, smoke-test it	Image size, container logs

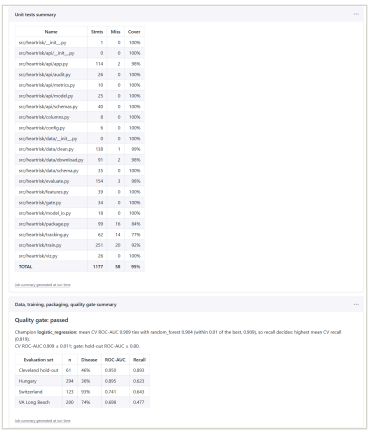
Each job has a timeout, runs bash with `-euo pipefail`, and uses the current major versions of the official actions; the workflow passes actionlint. The first run on `main` shows a failure being caught: the quality-gate step invoked the gate module without the package on its import path, so the job stopped with a clear `ModuleNotFoundError` in the log. Running the gate through `make` fixed it (commit `60f4cd9`), and the following runs were green.



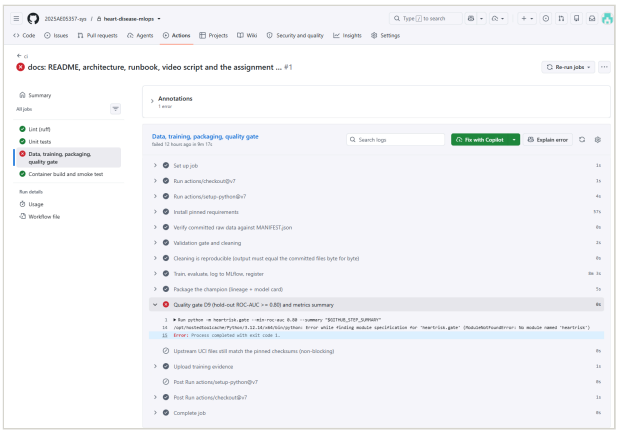
CI run with all jobs green



Artifacts uploaded by a CI run



Job summary with the quality gate and metrics



A failing quality-gate step stops the pipeline (run #1)

8. Serving and containerisation

The API (FastAPI) loads the model once at start-up and exposes:

Endpoint	Purpose
`POST /predict`, `POST /predict/batch`	Class, probability, confidence, risk band, imputed fields and the model's identity (batch: up to 100 patients)
`GET /model-info`	The model's `lineage.json`
`GET /health`, `GET /ready`	Liveness; readiness returns 503 until a model is loaded
`GET /metrics`, `GET /docs`	Prometheus series; interactive documentation

The input contract mirrors the data gate exactly (a test enforces it). Only age, sex and chest-pain type are required, because they are recorded at every site; any other value may be null, and the response lists it under `imputed_fields`, so the caller knows when an answer rests on estimates. Every response carries `X-Request-ID` (propagated or generated) and `X-Model-Version`. Every prediction writes one JSON audit line with the request id, model version, prediction and the SHA-256 of the canonical input, but not the clinical values themselves, so the log can prove which input produced which answer without holding patient data.

The image is multi-stage: wheels are built in a first stage and bind-mounted, not copied, into the runtime stage, which cut the download from 245 MB to 162 MB. It runs as a non-root user (uid 10001), has a health check, and carries the commit and model version as OCI labels; a `.dockerignore` allow-list sends only the serving lock, the package and the model to the build.

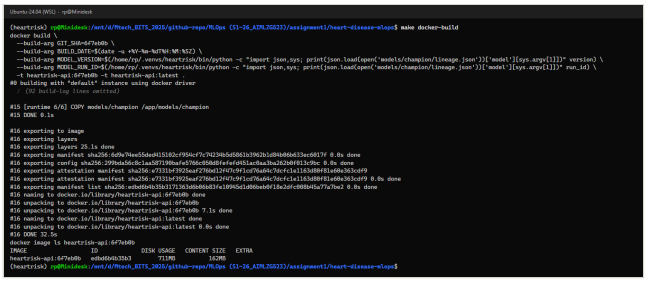
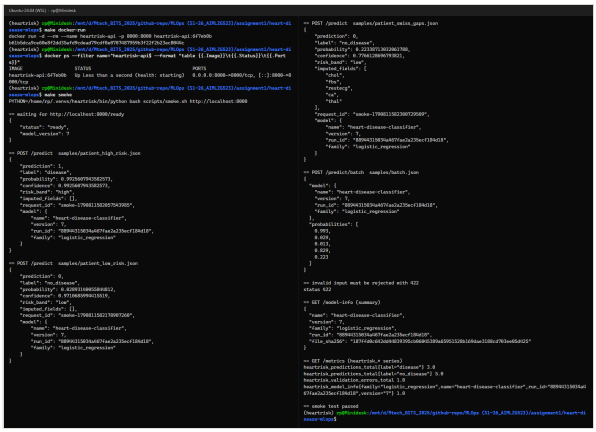
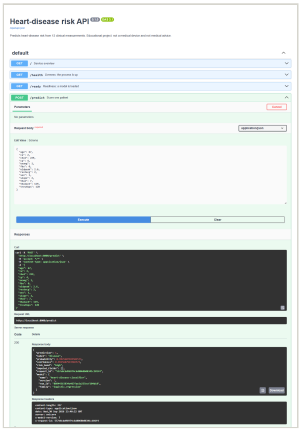


Image build



Smoke test against the container



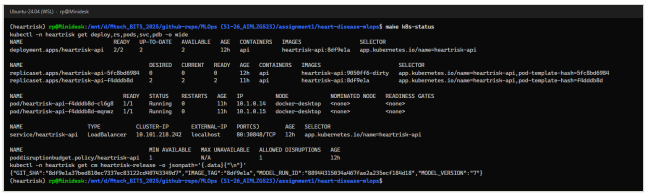
Interactive API documentation with an executed prediction

9. Deployment on Kubernetes

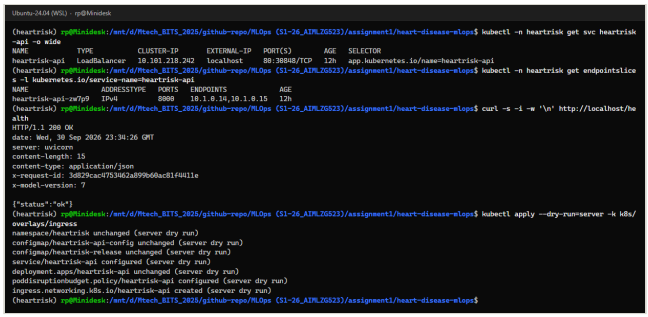
The manifests use kustomize: a base, a release component and two overlays.

Object	Key settings
Namespace	Enforces the **restricted** Pod Security Standard, so Kubernetes itself rejects any pod that runs as root, keeps capabilities or can escalate privileges
Deployment	2 replicas; rolling updates with `maxUnavailable: 0`; CPU and memory requests and limits; read-only root filesystem; no service-account token; Prometheus annotations
Probes	Startup and liveness on `/health`, readiness on `/ready`: a pod whose model fails to load stays out of the Service instead of crash-looping
Service	LoadBalancer on port 80 (Docker Desktop publishes it on localhost)
PodDisruptionBudget	At least one pod stays available during voluntary disruptions
Release component	`make k8s-release` writes image tag, commit, model version and run id; kustomize `replacements` stamp them onto the pod template and keep them in ConfigMap `heartrisk-release`

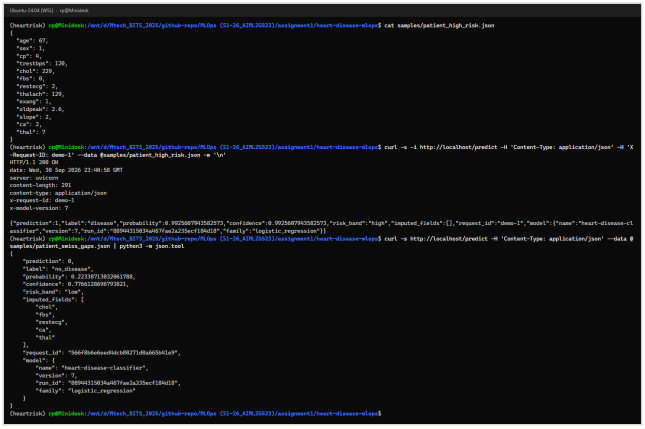
The brief allows a load balancer or an ingress (D11). The LoadBalancer overlay is deployed; an ingress overlay for Traefik at `heartrisk.localtest.me` passes a server-side dry run and is ready for when an ingress controller is installed. Deleting a pod demonstrates self-healing: the Deployment had a replacement pod Ready within about 20 seconds while the other pod kept serving.



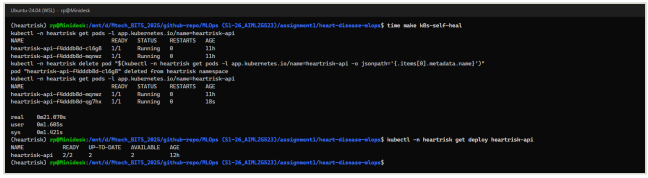
Pods, service and deployment in the cluster



Service exposure



Prediction through the load balancer



Pod deleted and replaced

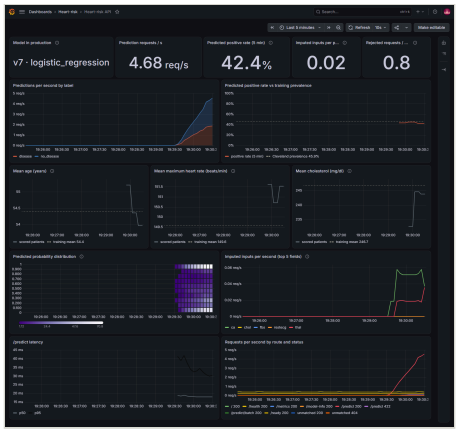
10. Monitoring and logging

Prometheus discovers the annotated API pods through a read-only role, scrapes them every 5 seconds and labels each series with the pod and the model version. The API exposes request counts and latency, predictions by label, the probability distribution, the distributions of age, cholesterol and maximum heart rate, imputed inputs per field, validation errors and the model's identity. Grafana is provisioned entirely from code, with its data source, a dashboard and an admin password from a Secret created at deploy time. Both run under the same restricted Pod Security Standard.

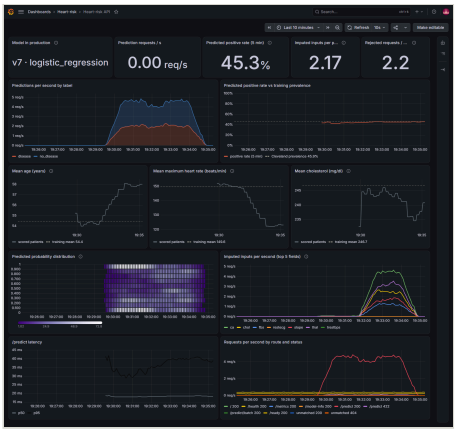
The external validation in Section 4 predicted how this model fails: on patients from another hospital. `scripts/traffic.py` replays real patients as traffic, from Cleveland in normal mode and from Switzerland and VA in shift mode, and the results confirmed the prediction:

Signal	Cleveland traffic	Swiss + VA traffic
Mean maximum heart rate	≈ 151 beats/min (training mean 149.6)	≈ 120 beats/min
Mean age	≈ 54 years (training mean 54.4)	≈ 58 years
Imputed inputs per prediction	≈ 0	≈ 3
Predicted positive rate	≈ 44%	≈ 47%, although 82% of these patients have disease
Alerts	none	`HeartriskInputDrift` and `HeartriskHighImputation` firing

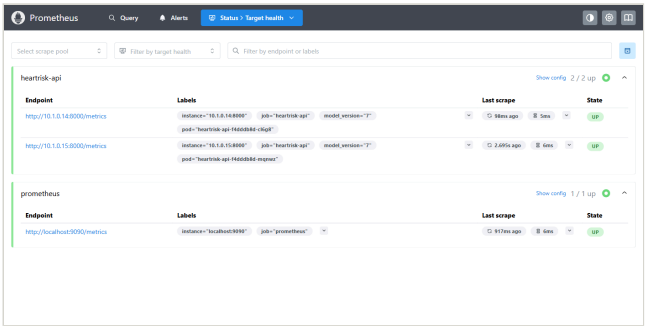
The last two rows are the most important result of the project. Under drift the model silently under-calls disease, and its output distribution barely moves, so an alert on the predicted positive rate alone stayed quiet. The alerts that fired watch the **inputs**: mean maximum heart rate more than 15 beats/min away from the training mean, and more than one imputed input per prediction. Logs are JSON lines on stdout (access and audit), read with `make k8s-logs`.



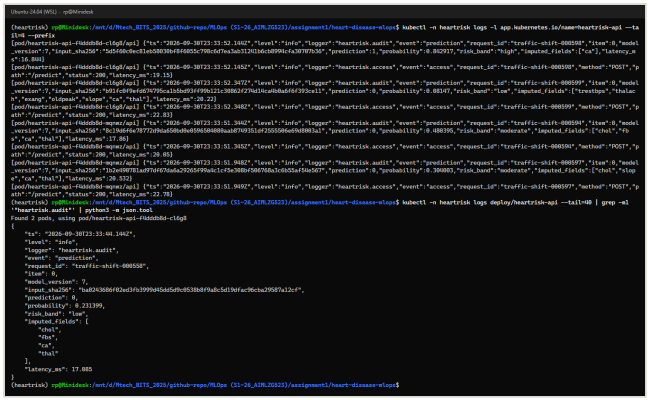
Grafana dashboard under normal traffic



Grafana dashboard during shifted traffic



Prometheus scrape targets



JSON access and audit logs

11. Results, limitations and future work

The champion reaches a cross-validated ROC-AUC of 0.909 and 0.950 on the hold-out set, but 0.70–0.74 at the VA and Swiss hospitals. The data explain why: those patients are older, reach lower heart rates and mostly lack the two strongest tests. The model card lists the limitations in full.

Limitation	Future work
242 training patients from the 1980s	Retrain on a larger, contemporary cohort; report confidence intervals
Poor transfer to other hospitals	Recalibrate per site, or train with site-aware methods once labelled local data exist
Probabilities calibrated to Cleveland's prevalence	Per-site recalibration; monitor calibration once outcomes arrive
Sex carries a large weight (cohort effect)	Fairness evaluation on larger slices
Drift is detected, not acted on	Route a retraining pipeline from the drift alert, with the challenger as a shadow model

12. Conclusion

The pipeline goes from four checksum-pinned public files to a monitored API on Kubernetes, and every link in between is recorded: a prediction names its model version, the model's lineage names its training run, commit and data hashes, and the deployment names its release. Traceability shaped the design choices, and it paid off in the monitoring stage, where tracing performance back to the data made the model's real weakness visible and turned it into an alert that fires.

Repository: https://github.com/2025AE05357-sys/heart-disease-mlops · Data: Janosi, Steinbrunn, Pfisterer & Detrano (1989), UCI Machine Learning Repository, https://doi.org/10.24432/C52P4X (CC BY 4.0).